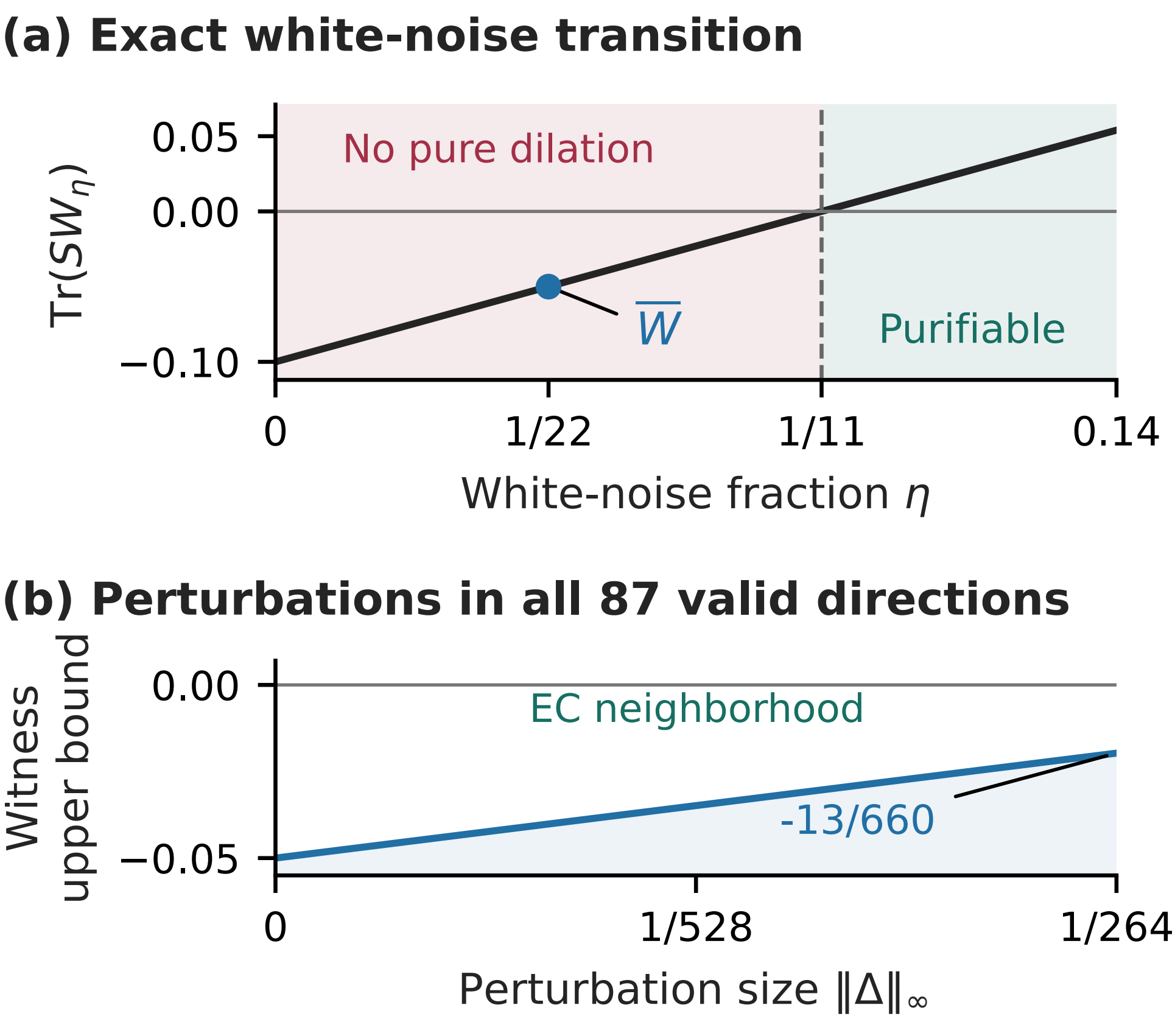
(a) Exact white-noise transition
0.05
0.00
−0.10
Tr(SW_η)
No pure dilation
W̄
Purifiable
0
1/22
1/11
0.14
White-noise fraction η
(b) Perturbations in all 87 valid directions
Witness upper bound
0.00
−0.05
EC neighborhood
-13/660
0
1/528
1/264
Perturbation size ‖Δ‖∞
Witness stays negative for every valid perturbation